Finite-Scale Response and Source-Centering Certificates for Weakly Interacting Bulk Ising Correlations

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Abstract

We examine the stated conformal-universality mechanism for weak, finite-range, spin-flip-even perturbations of the planar Ising model. Conditional on its analytic renormalization maps, thermal normalization, and uniform stable contraction, we derive finite-prefix certificates for both the slope and the curvature of the selected inverse-temperature branch. Ordered response operators account for stable-to-thermal leakage, while explicit geometric tails bound all omitted scales, including the quadratic forcing and initial mixed derivatives. A nearest-neighbor perturbation checks the factor $2Jr_0$ and the curvature convention. We also sharpen the common-continuation total-variation bound to an optimal nonlinear modulus and give a partition error certificate that removes only actual-state energy singleton cumulants. These are local quantitative consequences, not an independent proof of the source's renormalization construction, radial comparison, or conformal limit. Original local derivative and residual arguments are compared with the exact Ising interfaces, with no inference of infinite-dimensional control from finite tests. Contribution assessment: C3/5, explicit first- and second-response tail certificates and sharp finite source-error bounds provide a notable local increment under unchanged global analytic premises.

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1 Model, scope, and proof dependencies

Let $J > 0$ and let V be a fixed finite-range, translation-invariant, spin-flip-even interaction on $\mathbb{Z}^2$, invariant under reflections and quarter turns. With unordered nearest-neighbor bonds and unordered interaction sets, the source [1] uses the Hamiltonian

$$\mathcal{H}(\sigma) = -J \sum_{\{x,y\}\ \mathrm{nn}} \sigma_x \sigma_y - \lambda \sum_A V(A) \prod_{x \in A} \sigma_x.$$

The real parameter λ is small and fixed before the mesh tends to zero. At $\lambda = 0$, $\beta_0 = K_c/J$ with $K_c = \frac{1}{2}\log(1+\sqrt{2})$. The claimed result selects one analytic $\beta_c(\lambda)$ and analytic spin and energy amplitudes, independently of the domain and the fixed finite number of insertions. For bounded simply connected C^2 Jordan domains it treats free, fixed-plus, and fixed-minus boundary conditions and separated bulk insertions. The periodic thermodynamic limit precedes the mesh limit in the plane statement. These quantifiers are not statements about colliding points, arbitrary boundary insertions, interfaces, or all infinite-volume Gibbs states.

The scaled fields are $a^{-1/8}\sigma_{x_a}$ and $a^{-1}(\sigma_{x_a}\sigma_{y_a} - \mathbb{E}_{\mathrm{actual}}\sigma_{x_a}\sigma_{y_a})$. The latter expectation is in the interacting finite domain or the specified thermodynamic state, not automatically in the plane reference state. Fixed-boundary terms intersecting the domain include their prescribed outside spins; free-boundary terms must be contained in the domain.

1.1 Architecture and retained obligations

The following map records what the source actually uses. Reading a proof does not independently validate every imported estimate within it.

Source section	Mechanism and boundary of the present work
Inputs	Critical nearest-neighbor conditional laws, FK comparison, stopping-band resolution, primary-field limits, and fixed-profile exponential moments. The companion stopping theorem is an import.
Buffers	Common positive continuation tilts, ordered transmission, soft magnetic barriers, and two-sided truncation of normalized exponentials. The order of cutoffs and mesh limits matters.
Radial	Reflection quotient, dilation semigroup, charge identity, spectral gaps, and uniform bounded-test comparison. Uniformity does not follow just from convergence of finitely many point fields.
Renormalization	Protected carriers, connected logarithms and tree bounds, analytic polymer maps, exact thermal transport, stable quotient contraction, marked-source blocks, and boundary estimates.
Critical branch	Backward thermal and forward stable equations select an analytic decaying trajectory; an exact microscopic derivative selects the temperature by an implicit-function argument.
Correlations	Every source scalar is retained before centering; separated-source defects, domain restriction, torus winding errors, and local thermodynamic comparison complete the asserted limit.

The reference conditional law in this scheme remains the critical nearest-neighbor law. A finite Gram calculation or a coordinate Taylor expansion supplies neither its radial gap nor the volume-uniform polymer bounds. Our main theorem starts *after* those analytic interfaces have been obtained.

2 Ordered response operators

Fix real Banach spaces W_j and use the norm $\|(t,w)\| = |t| + \|w\|$ on $X_j = \mathbb{R} \oplus W_j$. Assume analytic maps $\mathcal{R}_j : X_j \to X_{j+1}$ on uniform neighborhoods of zero, with $\mathcal{R}_j(0) = 0$ and

$$T_j = D\mathcal{R}_j(0) = \begin{pmatrix} L & B_j \\ 0 & D_j \end{pmatrix}, \qquad L > 1, \quad 0 < \rho < 1, \quad \|D_j\| \le \rho, \quad \|B_j\| \le B. \tag{1}$$

Here $B_j : W_j \to \mathbb{R}$, $D_j : W_j \to W_{j+1}$; scalar constants B, H are upper bounds, not matrices. Also assume $\|Q_j\| \le H$ for the symmetric bilinear maps $Q_j = D^2\mathcal{R}_j(0)$, in the stated norms. These are hypotheses in the source's physical polymer coordinates, not consequences of changing to an arbitrary finite basis.

Let $h_0(\beta, \lambda) \in X_0$ be the analytic preliminary map. Assume an analytic branch $\beta_c(0) = \beta_0$ has a decaying orbit $h_j(\lambda)$, analytic as a sequence in a weighted norm $\sup_j \gamma^{-j} \|h_j\| < \infty$ for some $\rho < \gamma < 1$. In particular its first two derivatives at zero decay and may be differentiated through the recurrence. At $(\beta_0, 0)$ assume

$$h_0 = 0, \qquad h_\beta = (\tau, 0), \quad \tau > 0, \qquad h_\lambda = (a, b). \tag{2}$$

The source normalization is $\tau = 2Jr_0$ (critical.tex, lines 153–176). Indeed its site energy $e(z) = \frac{1}{4}\sum_{|w-z|=1} \sigma_z\sigma_w$ satisfies $\sum_z e(z) = \frac{1}{2}\sum_{\{z,w\}\ \mathrm{nn}} \sigma_z\sigma_w$. The extra factor two cannot be absorbed while keeping its thermal coordinate fixed.

For $i \ge j$ set $P_{j,j} = I$ and $P_{i,j} = D_{i-1} \cdots D_j : W_j \to W_i$. The order of multiplication is essential. Define

$$S_j = \sum_{i=j}^{\infty} L^{j-1-i} B_i P_{i,j}, \qquad s = \frac{B}{L - \rho}, \quad A = 1 + s, \quad r = \rho/L. \tag{3}$$

The series converges in operator norm and $\|S_j\| \le s$. For an integer $M \ge 1$, define $S_M^{[M]} = 0$ and

$$S_j^{[M]} = L^{-1}(B_j + S_{j+1}^{[M]} D_j), \quad 0 \le j < M. \tag{4}$$

Thus $S_j^{[M]}$ is the partial sum through $i = M - 1$ and

$$\left\| S_j - S_j^{[M]} \right\| \le s r^{M-j} \quad (0 \le j \le M). \tag{5}$$

This follows by summing $L^{-1}\sum_{k=M-j}^{\infty} (\rho/L)^k B$. There is no commutation or stationary-map assumption.

Theorem 2.1 (Slope and its finite-prefix certificate)**.** *Under the preceding hypotheses, writing* $\beta_1 = \beta_c'(0)$*,*

$$\beta_1 = -\frac{a + S_0 b}{\tau}, \qquad \beta_1^{[M]} = -\frac{a + S_0^{[M]} b}{\tau}, \qquad |\beta_1 - \beta_1^{[M]}| \le \delta_M := \frac{s\,\|b\|}{\tau} r^M. \tag{6}$$

Moreover the differentiated orbit $x_j = h_j'(0)$ *is exactly* $x_j = (-S_j w_j, w_j)$*,* $w_j = P_{j,0} b$*, so that* $\|x_j\| \le A\,\|b\|\,\rho^j$*.*

Proof. Differentiating gives $w_{j+1} = D_j w_j$ and $t_{j+1} = L t_j + B_j w_j$. Iterating the latter backwards from $m > j$,

$$t_j = L^{j-m} t_m - \sum_{i=j}^{m-1} L^{j-1-i} B_i w_i.$$

The first term tends to zero by the decaying-sequence hypothesis; absolute convergence gives $t_j = -S_j w_j$. Since $t_0 = \tau\beta_1 + a$, the formula and the tail bound follow. □

Even the linear response is generally not $-a/\tau$: the stable initial component b leaks into the thermal coordinate through all B_i. The source establishes qualitative analyticity; (6) makes that leakage and its finite cutoff explicit.

3 Curvature with every omitted scale bounded

All initial derivatives below are evaluated at $(\beta_0, 0)$. They are raw derivatives, not Taylor coefficients: Q_j is twice the quadratic Taylor coefficient. Define the vector polynomial

$$g_2(v) = h_{\lambda\lambda} + 2vh_{\beta\lambda} + v^2 h_{\beta\beta}. \tag{7}$$

Write $g_2(\beta_1) = (a_2, c_2)$ and $f_j = Q_j[x_j, x_j] = (f_j^t, f_j^w)$.

Theorem 3.1 (Exact curvature response)**.** *The second derivative $\beta_2 = \beta_c''(0)$ satisfies*

$$\beta_2 = -\frac{1}{\tau}\left[a_2 + S_0 c_2 + \sum_{j=0}^{\infty} L^{-j-1}(f_j^t + S_{j+1} f_j^w)\right]. \tag{8}$$

This series is absolutely convergent, with its tail from $j = M$ bounded in absolute value by

$$\frac{HA^3 \|b\|^2}{L - \rho^2}\left(\frac{\rho^2}{L}\right)^M \tag{9}$$

before division by τ.

Proof. For $y_j = h_j''(0)$, the differentiated recurrence is $y_{j+1} = T_j y_j + f_j$, with $y_0 = (\tau\beta_2 + a_2, c_2)$. The stable component is $y_j^w = P_{j,0} c_2 + \sum_{\ell<j} P_{j,\ell+1} f_\ell^w$. The backward thermal equation consequently gives

$$\tau\beta_2 + a_2 = -\sum_{j\geq 0} L^{-j-1}(B_j y_j^w + f_j^t).$$

Substituting the stable expression, its initial term is $-S_0 c_2$. For a fixed ℓ, the later leakage of f_ℓ^w is $-L^{-\ell-1} S_{\ell+1} f_\ell^w$. The double series is absolutely convergent because $\|f_j\| \leq HA^2 \|b\|^2 \rho^{2j}$ and $\|S_j\| \leq s$. This proves the identity and permits the interchange. The same bounds give $\sum_{j\geq M} L^{-j-1} AHA^2 \|b\|^2 \rho^{2j}$, which is (9). □

3.1 A computable-prefix formula, conditional on norm bounds

Use only B_j, D_j, Q_j for $0 \leq j < M$, the initial two-jet, and the declared global bounds. Set

$$\widehat{g}_2 = g_2(\beta_1^{[M]}) = (\widehat{a}_2, \widehat{c}_2), \quad \widehat{x}_j = (-S_j^{[M]} w_j, w_j), \quad \widehat{f}_j = Q_j[\widehat{x}_j, \widehat{x}_j].$$

Define

$$\beta_2^{[M]} = -\frac{1}{\tau}\left[\widehat{a}_2 + S_0^{[M]}\widehat{c}_2 + \sum_{j=0}^{M-1} L^{-j-1}(\widehat{f}_j^t + S_{j+1}^{[M]}\widehat{f}_j^w)\right]. \tag{10}$$

Evaluability means that the displayed finite operator actions and derivative bounds are supplied. It does not provide an oracle for infinite polymer spaces or certify the bounds from sampled entries.

Theorem 3.2 (Full curvature enclosure)**.** *For $M \geq 1$, let $b_0 = \|b\|$ and*

$$e_g = \delta_M \left(2\,\|h_{\beta\lambda}\| + (2|\beta_1^{[M]}| + \delta_M)\,\|h_{\beta\beta}\|\right).$$

Then

$$|\beta_2 - \beta_2^{[M]}| \leq \frac{1}{\tau}\Bigg[Ae_g + s\,\|\widehat{c}_2\|\,r^M + \frac{HA^3b_0^2}{L-\rho^2}(\rho^2/L)^M + \frac{HA^2sb_0^2}{L(1-\rho)}(2r^M + r^{M-1})\Bigg]. \tag{11}$$

Proof. The polynomial identity (7) and (6) give $\|g_2(\beta_1) - \widehat{g}_2\| \leq e_g$. The functional $(t, w) \mapsto t + S_0 w$ has norm at most A, so the initial discrepancy is at most $Ae_g + s\,\|\widehat{c}_2\|\,r^M$. Equation (9) bounds the missing infinite part. For $j < M$, both x_j and $\widehat{x}_j$ have norm at most $Ab_0\rho^j$ and

$$\|x_j - \widehat{x}_j\| \leq sb_0\rho^j r^{M-j}, \qquad \left\|f_j - \widehat{f}_j\right\| \leq 2HAsb_0^2\rho^{2j}r^{M-j}.$$

Applying the response functional costs at most A. Consequently the sum of these errors is at most $2HA^2sb_0^2r^M/[L(1-\rho)]$, since $L^{-j}\rho^{2j}r^{-j} = \rho^j$. Replacing the future response S_{j+1} by $S_{j+1}^{[M]}$ in the remaining finite terms costs at most $HA^2sb_0^2r^{M-1}/[L(1-\rho)]$, by the same summation. Adding the four contributions proves the assertion. □

All terms are necessary types of error, though the constants are conservative. In particular, omitting $h_{\beta\lambda}$ or the stable component f_j^w generally gives an incorrect curvature. A finite matrix truncation of W_j would require additional projection and operator-action errors; the omitted-scale bound is not such a certificate.

3.2 Two normalization checks

First take a perturbation supported only on nearest-neighbor pairs, with coefficient v on every unordered bond. Then the model is exactly the nearest-neighbor model with coupling $\beta(J + \lambda v)$, and

$$\beta_c(\lambda) = \frac{K_c}{J + \lambda v}, \quad \beta_1 = -\frac{\beta_0 v}{J}, \quad \beta_2 = \frac{2\beta_0 v^2}{J^2}. \tag{12}$$

The preliminary map is a function $F(q)$ of $q = \beta(J + \lambda v) - K_c$, with $F'(0) = 2r_0(1, 0)$. Thus $b = 0$, $a = 2\beta_0 v r_0$, and (6) reproduces the first derivative. In $g_2(\beta_1)$ all $F''(0)$ terms multiply $(\beta_0 v + J\beta_1)^2 = 0$; the remainder is $2\beta_1 v F'(0)$. Consequently $c_2 = 0$, $a_2 = -4\beta_0 v^2 r_0/J$, and (8) gives the second derivative in (12). Here $x_j = 0$ and every nonlinear forcing vanishes. Keeping only $h_{\lambda\lambda}$ would fail this check.

For a genuinely nonzero stable direction, the algebraic model

$$\mathcal{R}(t, w) = (Lt + Bw + qw^2, dw), \qquad |d| < 1 < L,$$

has the exact stable graph $t = -Bw/(L-d) - qw^2/(L-d^2)$. With $h_0 = (\tau(\beta - \beta_0) + a\lambda + c\lambda^2/2, b\lambda)$,

$$\beta_1 = -\frac{a + Bb/(L-d)}{\tau}, \qquad \beta_2 = -\frac{c + 2qb^2/(L-d^2)}{\tau}.$$

The forcing in (8) is $2qb^2d^{2j}$, not qb^2d^{2j}. This is an exact fixture, not an Ising renormalization map or numerical evidence for its global bounds.

4 Optimal common-continuation error transfer

For probabilities μ, ν on a common measurable space use $\mathrm{TV}(\mu,\nu) = \sup_A |\mu(A) - \nu(A)|$. Let w be a common positive measurable weight whose essential ratio on the union of the supports is at most $K \geq 1$. Write $\mu^w = w\mu/\mu(w)$ and similarly for ν. In the source's buffer lemma this is the ratio of two outer partition functions after inner data have been fixed. A different weight for each inner law would not meet the following hypothesis.

Theorem 4.1 (Sharp common-tilt modulus)**.** *If $d = \mathrm{TV}(\mu,\nu)$, then*

$$\mathrm{TV}(\mu^w, \nu^w) \leq F_K(d) := \frac{Kd}{1 + (K-1)d}. \tag{13}$$

The bound is optimal for every $K \geq 1$ and $d \in [0,1]$. It improves the source estimate $2Kd$, and is at most Kd.

Proof. Scale w so $1 \leq w \leq K$; limiting essential bounds give the same conclusion if the extrema are not attained. With dominating measure $\mu + \nu$, the common submeasure $\zeta = \mu \wedge \nu$ has mass $1 - d$. Write $\mu = \zeta + \alpha$, $\nu = \zeta + \beta$, with $\alpha(\Omega) = \beta(\Omega) = d$. Let $c = \zeta(w)$, $u = \alpha(w)$, and $v = \beta(w)$. The normalized laws share at least the submeasure $w\zeta/\max(c+u, c+v)$. Their overlap is therefore at least $c/(c + \max(u,v)) \geq (1-d)/(1-d+Kd)$. Subtracting from one proves (13), including the endpoint cases directly. Equality holds on three atoms for $\mu = (1-d, d, 0)$, $\nu = (1-d, 0, d)$, $w = (1, K, K)$. □

For example $K = 3, d = 1/10$ gives $1/4$, whereas $2Kd = 3/5$. A common Markov kernel applied afterwards contracts total variation: for $0 \leq f \leq 1$, integration against $\mu - \nu$ is bounded by d, so apply this to the kernel probability of an event. No independence of successive annuli is required. Moreover $F_K(F_H(d)) = F_{KH}(d)$ by direct substitution. To guarantee a final tolerance $\epsilon < 1$, it suffices and, in the worst case, is necessary that

$$d \leq \frac{\epsilon}{K - (K-1)\epsilon}.$$

This composition statement concerns only the specified common tilts. Inserted approximation errors, changes of kernels, and exceptional transcripts must be charged separately.

For example, suppose the reference stopping construction has good-law distance at most η, its bad event has probability at most ε, and a separate pointwise continuation estimate bounds the target bad-event probability by $C\varepsilon$. Then the target bounded-test comparison may use $2F_K(\eta) + 2C\varepsilon$ when the exceptional comparison is charged by twice its probability. The factors C and K require the source's genuine geometric comparison. This elementary improvement does not construct the stopping band or make those constants uniform over arbitrary geometries.

5 Normalization, derivatives, and actual-state centering

5.1 Differentiating a normalized law

On a finite probability space with fixed reference law μ, let $\mu_\theta = e^{H_\theta}\mu/\mu(e^{H_\theta})$ for real C^2 H_θ. For a real C^2 observable A_θ, direct differentiation gives

$$\frac{d}{d\theta}\mathbb{E}_\theta A_\theta = \mathbb{E}_\theta A'_\theta + \mathrm{Cov}_\theta(A_\theta, H'_\theta), \tag{14}$$

$$\begin{aligned}\frac{d^2}{d\theta^2}\mathbb{E}_\theta A_\theta &= \mathbb{E}_\theta A''_\theta + 2\,\mathrm{Cov}_\theta(A'_\theta, H'_\theta) + \mathrm{Cov}_\theta(A_\theta, H''_\theta)\\ &\quad + \mathbb{E}_\theta\big[(A_\theta - \mathbb{E}_\theta A_\theta)(H'_\theta - \mathbb{E}_\theta H'_\theta)^2\big].\end{aligned} \tag{15}$$

To prove the second identity, apply the first to each of $\mathbb{E}A'$, $\mathbb{E}(AH')$, and $(\mathbb{E}A)(\mathbb{E}H')$ in the first identity. The terms containing three factors combine into the displayed centered third cumulant; the remaining terms have the stated coefficients. This proves the formula without omitting the denominator derivative. Adding a scalar function to H_θ changes neither formula. For an infinite space the same calculation requires domination of the displayed derivatives; it is not inferred from the finite version.

In a fixed Ising volume, let $\mathcal{B}$ be the unordered bond sum and $\mathcal{V}$ the interaction sum. Along a selected branch the relative exponent is $H_\lambda = (J\beta_c(\lambda) - K_c)\mathcal{B} + \lambda\beta_c(\lambda)\mathcal{V}$. At zero,

$$H'_0 = J\beta_1\mathcal{B} + \beta_0\mathcal{V}, \qquad H''_0 = J\beta_2\mathcal{B} + 2\beta_1\mathcal{V}.$$

Thus even fixed-volume second response requires the mixed temperature term. Finite-volume real normalization is positive; this is not a uniform complex zero-free result or a mesh-uniform susceptibility bound.

5.2 A finite joint-source certificate

Work in the commutative nilpotent algebra $\mathcal{A}_q = \mathbb{C}[t_1, \dots, t_q]/(t_1^2, \dots, t_q^2)$. For observables $O_1, \dots, O_q$ in one fixed actual state, let

$$G(t) = \mathbb{E}\prod_i (1 + t_i O_i), \qquad K(t) = \log G(t) = \sum_{\emptyset \neq S \subseteq [q]} \kappa_S t_S.$$

These are ordinary mixed cumulants; distinct labels have no factorial normalization. Let $I_\varepsilon \subseteq [q]$ be the energy labels. Define $QK = K - \sum_{i \in I_\varepsilon} \kappa_{\{i\}} t_i$. Then exact centering of those energy observables changes G to $\exp(QK)$. In particular it does not remove spin singleton coefficients.

Proposition 5.1 (Partition certificate with the correct deleted terms)**.** *Let Π_ε be the set of partitions of $[q]$ having no singleton block $\{i\}$ with $i \in I_\varepsilon$. Then*

$$[t_1 \cdots t_q]\exp(QK) = \sum_{\pi \in \Pi_\varepsilon} \prod_{S \in \pi} \kappa_S. \tag{16}$$

If retained coefficients obey $|\widehat{\kappa}_S| \le M_S$ and $|\kappa_S - \widehat{\kappa}_S| \le e_S$, the error on replacing all coefficients by their approximations is at most

$$\sum_{\pi \in \Pi_\varepsilon} \left\{ \prod_{S \in \pi} (M_S + e_S) - \prod_{S \in \pi} M_S \right\}. \tag{17}$$

Proof. In the exponential series only disjoint nonempty label sets can contribute to the full squarefree monomial. A partition with k blocks occurs $k!$ times, cancelling the exponential's $1/k!$. Deleting the specified singleton terms gives exactly Π_ε. In each product expand $\kappa_S = \widehat{\kappa}_S + (\kappa_S - \widehat{\kappa}_S)$. The sum over all nonempty choices of error factors is bounded by the difference of products in (17). □

For one spin and two energies this reduces to $\kappa_{123}+\kappa_1\kappa_{23}$, with error bound $e_{123}+M_1e_{23}+M_{23}e_1+e_1e_{23}$. Deleting the spin singleton would lose a genuine term. If the source has already extracted a scalar c_i at an energy label, then $\kappa_i = c_i + \mathbb{E}_{\text{residual}}U_i$; both terms must be subtracted. A plane mean substituted for this actual residual mean would implement a different observable. Scalar coefficients involving two or more labels remain in (16), even for energy labels. The formula assumes exact own-state centering; deliberately approximate centering requires charging its remaining singleton errors separately.

6 Separated marks and the limits of finite certificates

The source obtains defects of the form $\epsilon_j \exp(-c\Delta/r_{j+1})$ with bounded $\epsilon_j \to 0$, after subtracting the ideal separated-mark evolution. At a fixed stopping fraction r_k/Δ, the spatial exponential at the last scale is a fixed small number, not a sequence tending to zero. Vanishing of the defect must also use the decaying background and marked remainder.

More explicitly, if $K_* > 0$, $L > 1$, and $c_* > 0$ are fixed, then

$$\sum_{h=0}^{k-1} K_*^h e^{-c_* L^h} \epsilon_{k-h-1} \longrightarrow 0.$$

For each fixed h the summand tends to zero. It is dominated by a constant times $K_*^h e^{-c_* L^h}$, whose series converges by the ratio test. This proves the assertion. If instead $\epsilon_j = 1$, the $h = 0$ term is e^{-c_*} and does not vanish. This simple check prevents a false upgrade from localization alone.

The scale cutoff in Theorem 3.2 is different: it controls derivatives at the unperturbed branch under uniform bounds. It gives no rate in mesh for arbitrary fixed λ. Neither it nor Proposition 5.1 proves that source cumulant errors tend to zero; such errors still have to be supplied by the analytic maps and radial transfer. The exact disappearance of an ideal connected coefficient for disjoint protective buffers is also not a factorization of unconditional critical correlations.

7 Original-method comparison and present literature

The fresh original-corpus comparison is deliberately narrower than the global Ising problem. The original TR6 derivations [6], lines 335–425, differentiate nonlinear and normalized integral operators; the normalized derivative retains its denominator contribution. Its local contraction argument, lines 657–680, gives an error bound only on a verified ball. The original TR1 stability chapter [7] distinguishes finite perturbations from a full residual and requires control of unrepresented directions. The local-moment chapter [8], lines 1–63 and 163–225, separates Taylor remainder, constant reproduction, mass conservation, and true finite closure.

Here those methods lead to three concrete safeguards. Differentiating the initial microscopic map must retain (7); changing a normalized source requires (14)–(15); and finite scale truncation must carry (11), not just an invertible finite matrix. None of these private originals supplies critical Ising radial spectrum, FK stopping estimates, or an analytic polymer construction. Their inclusion is a comparison of actual proof mechanisms, not a claim that the present applications were already proved there.

The October 9, 2026 public-primary refresh distinguishes several scopes. Antinucci–Giuliani–Greenblatt [2] concerns energy correlations of weakly non-integrable Ising models on a cylinder; the primary abstract identifies the critical scaling limit and multiplicative normalization. Cava–Giuliani–Greenblatt [3] concerns *boundary* spin correlations in the half-plane and a Pfaffian limit, not the entire bulk mixed-field statement discussed here. Chelkak–Hongler–Izyurov [4] treats primary fields of the critical nearest-neighbor Ising model in finitely connected domains. Giuliani's review [5] provides the non-integrable renormalization context. These primary abstracts and revision metadata were inspected, not their complete proofs. They are useful scope checks, not a declaration of an exhaustive literature census or independent confirmation of the new cached manuscript.

7.1 Increment, advantages, and costs

The source already has the triangular block, a decaying analytic trajectory, qualitative analytic amplitudes, and exact joint-source bookkeeping. We do not count those as new. The local increment is the explicit two-jet response certificate: ordered S_j, both forcing channels, all initial mixed derivatives, and a finite-prefix curvature enclosure with an independent exact normalization example. It uses classical chain-rule and geometric-series arguments but answers a specific quantitative question that qualitative analyticity alone does not answer. The optimal common-tilt modulus and admissible partition error polynomial strengthen two finite error interfaces; their classical overlap and cumulant ingredients are credited as such. This supports C3 as a notable local improvement, not C4 or a global universality certification; applicability L3 is a separate classification.

An advantage is that the bounds allow different stable spaces at each scale and noncommuting maps. They expose which operator actions need rigorous approximation. For a desired slope tolerance ϵ, (6) gives an explicit sufficient cutoff once $s, \|b\|, \tau, r$ are known. The curvature bound similarly decreases geometrically for fixed constants. Its costs include full-space norm bounds, an initial second jet, and M applications of quadratic derivatives; there is no asserted computational complexity reduction for evaluating a polymer map. Bounds worsen as $\rho \uparrow 1$, and no uniformity over arbitrary perturbation ranges or geometries is asserted.

Future useful work would enclose these specific operator actions in the source's carrier norm, with projection residuals tracked separately. Higher-order response would require controlled higher derivatives and a majorant for the resulting partitions, not formal differentiation alone. A distinct problem is obtaining effective mesh thresholds for the buffered reference estimates; none of the present derivative cutoffs addresses it. No overlap with a previous report's determinant, kernel, or packing estimate is claimed as a new result here.

8 Verification and reading boundary

The owned script `M403-checks.py` performs exact rational arithmetic and passed 7,371 parameter cases comprising 22,225 assertions:

Fixture family	Cases	Assertions
Three-atom common tilts	6,075	18,225
Sharpness and composition	176	352
Quadratic invariant-graph identities	144	432
Full slope and curvature enclosures	720	2,880
Ordered noncommuting stable products	96	96
Actual-state partition centering	80	80
Normalized first and second derivatives	80	160
Total	7,371	22,225

The curvature fixtures compare the response expression with an independently solved quadratic graph equation, allowing quadratic forcing in both coordinates and mixed initial derivatives. They test signs, order, factors of two, and the declared inequalities. They are not evaluations of actual Ising RG maps. Parent-run fixtures are excluded from these totals.

All 12 cached public texts were displayed literally in bounded chunks: 4,224 lines, including all 3,883 lines in the eight scientific sections. The source's referenced companion and external proof literature remain imported; textual reading of the TikZ material is not a rendered-source-figure audit. Fresh private reads were TR6 266–453 and 571–722, TR1 stability 1–295, and the local-moment chapter 1–240, totaling 875 lines. Unread complements are recorded per file; included subsidiary files of the book chapter were not read. The pre-read receipt was persisted before these fresh literal reads. Post-read hashes and final artifact hashes are recorded separately. This is neither a whole-corpus reading nor a claim of an independent full-proof audit of the conformal theorem.

References

[1] *Conformal universality of bulk Ising correlations under weak interactions*, September 23, 2026, cached public manuscript. Scientific locators: buffers.tex 57–114; renormalization.tex and critical.tex for the analytic block and thermal normalization; correlations.tex for scalar extraction, centering, and separated defects. Pinned public source tree.

[2] G. Antinucci, A. Giuliani, R. L. Greenblatt, *Energy correlations of non-integrable Ising models: the scaling limit in the cylinder*, Communications in Mathematical Physics 397 (2023), 393–483. Primary arXiv record, v4.

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